Piyush Kumar Prajapati

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Areas of Interest

Quantitative MRI • Physics Informed Neural Networks • Medical Image Analysis • Deep Learning • Computer Vision • Generative AI and Diffusion Models • Uncertainty Quantification • Image Segmentation • AI in Healthcare

Education

Indian Institute of Technology Delhi (IIT Delhi) — New Delhi, India
Ph.D. in Biomedical Engineering, Centre for Biomedical Engineering; CGPA: 9.17 / 10 — *Sep. 2020 – Present*

- **Thesis:** Quantitative MRI and its Applications. **Advisor:** Prof. Anup Singh, MedImg Lab.
- Developed deep learning methods for fast, reliable quantitative DCE MRI in brain tumors, including CNN estimators, an AIF free three point framework and physics separable networks with probabilistic and diffusion posteriors. On grade 2 to 4 gliomas they ran about 66 times faster than conventional fitting with similar grading accuracy, calibrated uncertainty and robustness at a quarter of the standard contrast dose.

Indian Institute of Technology Ropar (IIT Ropar) — Rupnagar, India
M.Tech. in Biomedical Engineering, Center for Biomedical Engineering; CGPA: 9.02 / 10 — *Jul. 2018 – Jun. 2020*

- Computational Biomedical Imaging Lab. **Advisor:** Dr. Deepti R. Bathula.

Ajay Kumar Garg Engineering College (AKTU) — Ghaziabad, India
B.Tech. in Electrical and Electronics Engineering; Aggregate: 77.46% — *Aug. 2012 – Jul. 2016*

- **Project Advisor:** Rahul Dixit, Scientific Officer.

City Montessori School, Rajajipuram (ICSE) — Lucknow, India
Class XII: 95.16%; Class X: 90.43% — *2010 – 2012*

Publications

(links to the paper)

- [**MICCAI 2026**] **P. K. Prajapati**, M. T. Sheikh, A. Choudhary, M. Parikh, A. Singh. *"RACER: A Radiomic-Adaptive Cohort Ensemble with Refinement for Cross-cohort Brain Tumor Segmentation using Deep Learning."* BraTS Generalizability Across Tumors Challenge, MICCAI 2026 Satellite Events, Strasbourg, France.
- [**Magnetic Resonance Imaging 2026**] **P. K. Prajapati**, R. K. Gupta, A. Singh. *"Deep Learning Approach for DCE-MRI Parameter Estimation: Evaluating Signal Intensity and Concentration-Time Curve-Based Convolution Neural Networks."* Vol. 126, 110559.
- [**Under Review**] M. T. Sheikh, A. Kandpal, V. K. Yadav, **P. K. Prajapati**, R. K. Singh, R. Patir, S. Vaishya, S. Ahlawat, R. K. Gupta, A. Singh. *"Multiparametric MRI-based Longitudinal Assessment of Residual Tumor Volume and Microvascular Changes Following Glioblastoma Resection."*
- [**Ready for Submission**] **P. K. Prajapati**, A. Kandpal, S. Srivastava, R. Patir, M. C. Sharma, S. Vaishya, R. K. Singh, R. K. Gupta, A. Singh. *"Non-invasive Mapping of Tumor Cell Infiltration in the Peritumoral Region of Glioblastoma using Quantitative Dynamic Contrast Enhanced Perfusion MRI."*
- [**In Preparation**] **P. K. Prajapati**, A. Kandpal, M. T. Sheikh, R. K. Singh, R. K. Gupta, A. Singh. *"Dose-Robust and Acquisition-Independent Deep Learning for DCE-MRI Tracer-Kinetic Parameter Estimation: Application in Glioma."*
- [**IJSE 2015**] **P. Prajapati**, L. Kumar. *"Application Based Study of the Architecture of a SCADA System and Operational Stages."* International Journal of Science and Engineering, Vol. 3, pp. 137–141 (presented at ICAESM 2015, Agra).

Conferences

- [**ISMRM 2027, Submitted**] **P. K. Prajapati**, A. Kandpal, M. T. Sheikh, R. K. Singh, R. K. Gupta, A. Singh. *"ProTK-Net: A Probabilistic, Physics-Separable Network for Tracer-Kinetic Parameter Estimation in Glioma DCE-MRI."* Vancouver, Canada.
- [**ISMRM 2027, Submitted**] **P. K. Prajapati**, A. Kandpal, R. K. Singh, R. K. Gupta, A. Singh. *"Less Gadolinium, Same Tracer Kinetic Maps: A Generative, Physics Separable Estimator for DCE-MRI in Brain Tumor Patients."* Vancouver, Canada.
- [**ISMRM 2025**] **P. K. Prajapati**, A. Kandpal, S. Srivastava, R. K. Gupta, A. Singh. *"Three-Point Deep Learning Framework for Protocol-Independent and AIF-Free DCE-MRI Parameter Estimation in Gliomas."* Honolulu, USA. Digital Poster.

- [**ISMRM 2025**] A. Kandpal, **P. K. Prajapati**, S. Maurya, R. K. Gupta, A. Singh. *"A Two Step Deep Learning Framework for Identifying Ischemic Stroke Core: Integration of Inception-v3 and MultiResU-Net on DWI and ADC MRI Images."* Honolulu, USA. Digital Poster.
- [**ISMRM 2024**] **P. K. Prajapati**, A. Kandpal, R. Jafari, R. K. Gupta, A. Singh. *"Quantitative Analysis of DCE-MRI Data Using DL Model Based on Signal Intensity vs Concentration Curves."* Singapore. Digital Poster.
- [**ISMRM Workshop 2024**] **P. K. Prajapati**, A. Kandpal, R. K. Gupta, A. Singh. *"Denoising DCE-MRI Data Using Nadaraya-Watson Method: A Time-Efficient Approach for Improved Tracer Kinetic Maps."* 2nd ISMRM Workshop on Accessible MRI, New Delhi, India. Poster.
- [**ISMRM 2023**] **P. K. Prajapati**, R. K. Gupta, A. Singh. *"Quantitative DCE-MRI Parameter Estimation Using Deep Learning Framework in Brain Tumor Patients."* Toronto, Canada. Digital Poster.
- [**ASMRM 2023**] **P. K. Prajapati**, R. K. Gupta, A. Singh. *"Deep Learning Framework to Denoise DCE MRI Data."* 5th Annual Scientific Meeting of the Asian Society of Magnetic Resonance in Medicine. Poster.

Internships

CARING Research, Mahajan Imaging May 2019 – Aug. 2019
Research Intern: Knee Cartilage Segmentation from MRI *New Delhi, India*

- Developed a Python pipeline for atlas registration and knee cartilage segmentation using Elastix, achieving a Dice score of 0.94 on 20 knee MRI scans.
- Automated cartilage thickness estimation, cutting segmentation and analysis time from about 40 minutes to 5 minutes per scan.
- Implemented radiology report transcription with the Google Speech API, piloted on 30 audio reports, and contributed to workflow improvements and fMRI preprocessing.

Bharat Heavy Electricals Limited (BHEL), Insulator Plant Jun. 2015 – Jul. 2015
Student Intern: Insulator Manufacturing, Assembly and Testing *Jagdishpur, India*

- Studied 5 key stages of insulator manufacturing, assembly and testing, along with operations at a 33/11 kV power distribution substation.
- Identified 2 recurring production issues, surface defects and dimensional inconsistencies, and proposed better quality inspection, kiln temperature monitoring and mould calibration.

UNICON PLC Automation Jun. 2014 – Jul. 2014
Student Intern: Automatic Bottle Filling Plant using PLC and SCADA *Lucknow, India*

- Designed a prototype automatic bottle filling system on an Allen Bradley MicroLogix 1000 PLC, integrating 3 sensors, 2 actuators and 8 digital I/O points.
- Developed ladder logic and a Wonderware InTouch SCADA interface with automatic and manual modes, targeting 6 bottles per minute at $\pm 5\%$ volumetric accuracy.

Projects

Colorectal Cancer Detection and Segmentation | *M.Tech. Thesis, IIT Ropar* Jul. 2019 – May 2020

- Developed a two stage deep learning framework for automated colorectal cancer detection and segmentation on whole slide images (WSIs) from the DigestPath 2019 dataset, using VGG16 to classify histopathology patches as cancerous or non-cancerous.
- Integrated MobileNet with a segmentation decoder for pixel level tumor segmentation, with automatic bounding box extraction from the predicted masks to localise malignant regions.
- Designed a patch based processing pipeline for efficient analysis of high resolution WSIs, reaching a classification AUC of 0.94, Dice of 0.76, segmentation IoU of 0.69 and bounding box IoU of 0.78.

Single Image Super Resolution using Median Filter Transform | *CSL715, IIT Ropar* Jan. 2019 – May 2019

- Implemented median filter transform based super resolution in MATLAB to restore detail from a single noisy image, improving SSIM and other restoration measures.

Low Cost 3D Printed Syringe Pump | *BM608, IIT Ropar* Jan. 2019 – May 2019

- Designed and 3D printed a PLA syringe pump driven by a stepper motor and rack and pinion under Arduino Uno control, delivering 30 to 53 ml per hour as an economical alternative to commercial pumps.

Sleep Arousal Detection from SaO2 Signals | *BM605, IIT Ropar* Jul. 2018 – Dec. 2018

- Built a logistic regression model in MATLAB with multi window features from over 300 GB of PhysioNet Challenge 2018 data, achieving an AUC of 0.65 on held out patients.

Automatic Drilling of Pump Casing | *B.Tech. Final Year Project, AKGEC* Jun. 2015 – May 2016

- Built a prototype automatic pneumatic drill on an Allen Bradley MicroLogix 1000 PLC with integrated sensors, enabling accurate, programmable drilling around a pump casing model.

SCADA based Power Distribution and Theft Detection | *EEE502, AKGEC* Aug. 2014 – Dec. 2014

- Built an ATmega8535 based power distribution controller with a DS1307 real time clock for theft detection and remote monitoring; extended into a SCADA architecture study published in IJSE.

Technical Skills

Medical Imaging: DCE MRI, DWI and ADC, Tracer Kinetic Modeling, Variable Projection, Image Segmentation, Denoising, Digital Phantoms
Deep Learning: PyTorch, CNN, U-Net, nnU-Net, GRU with Attention, Diffusion Models, Simulation Based Inference, Transfer Learning
Programming: Python, MATLAB, SQL, LaTeX, Git
Data and Scientific Computing: NumPy, Pandas, Scikit-learn, Matplotlib

Teaching Assistantship

Teaching | *IIT Delhi* 2022 – Present

- Teaching Assistant for graduate courses in Deep Learning for Medical Image Analysis (BML7381, BML815B), Medical Imaging (BML720), Biomedical Data Analysis (BML738) and Biomedical Signal and Image Processing (BML735) across 6 semesters, including support for lectures, hands-on practicals and tutorials, assignments, examinations and student projects.
- Delivered sessions on ResNet and residual learning, support vector machines, logistic regression, quantitative imaging, and classical tumor segmentation in multiparametric MRI, from Otsu thresholding to active contours.
- Delivered sessions on sequence modelling, from RNNs to LSTM variants, for healthcare and medical professionals in the CEP Executive Programme for AI in Healthcare (2025 – 2026), and mentored two teams of 14 participants each through their capstone projects.

Teaching | *IIT Ropar* 2019 – 2020

- Lab Teaching Assistant for Physics of Medical Imaging (BM609), conducting ultrasound experiments on A, B and M mode imaging, Doppler flow measurement and ultrasound computed tomography on phantoms.

Positions of Responsibility

CBME Placement Coordinator, CDCRC | *IIT Ropar* Jul. 2019 – Jun. 2020
- Organised and coordinated placement activities for all students, while ensuring CBME students received fair representation and strong opportunities with recruiters.

Secretary, IEEE EMBS Student Chapter | *IIT Ropar* Jul. 2019 – Jun. 2020
- Coordinated chapter activities, meetings and member communication, and organised technical events for biomedical engineering students.

Vice President, Pehchaan Ek Safar NGO | *Government of Punjab approved NGO* Jul. 2019 – Jun. 2020
- Raised over Rs 2.5 lakh through fundraising campaigns, organised interactive teaching sessions for underprivileged children aged 5 to 18, and led a COVID drive collecting laptops and mobile phones for online learning.

Mess Representative, Hostel Mess Committee | *IIT Ropar* Jul. 2019 – Jun. 2020
- Represented hostel residents on mess matters, collecting feedback on food quality and menus and working with the committee and vendor to resolve issues.

Mentor, Movie Screening Club, Board of Literary Affairs | *IIT Ropar* Jul. 2019 – Jun. 2020
- Guided club members in planning and running regular movie screenings and discussions for the campus community.

Team Lead, Project KHEL NGO | *Lucknow* Jan. 2017 – Jul. 2018
- Led sport based life skills sessions for over 200 underprivileged children across 5 locations in Lucknow.

Vice Captain, EEE Department Sports Team, SHAKSHAM 2015 | *AKGEC Ghaziabad* Jun. 2015
Sports Captain, School Sports | *City Montessori School, Lucknow* Apr. 2011 – May 2012

Extra Curricular Activities

- **2nd Position**, Overall Championship, Men's General Championship and Water Polo, 58th Inter IIT Sports Meet, 2025.
- **Third Prize** (2015) and **Popular Choice Award** (2014), Anti Ragging Poster Making; **Second Position**, Laugh A Quake, Vibrations 2013, AKGEC.
- **Member**, College Soccer Team, AKGEC (2012 – 2015); **Student Member**, ISTE (2013 – 2016); **Volunteer**, World Welfare Organization, Rae Bareli (2012).
- **Silver Medal**, 100 m Butterfly Stroke, ASISC National Swimming Championship, 2011.

References

Prof. Anup Singh: PhD Supervisor, CBME and Yardi School of AI, IIT Delhi; AIIMS Delhi, anupsm@iitd.ac.in
Prof. Amit Mehndiratta: CBME and Yardi School of AI, IIT Delhi; AIIMS Delhi, amehndiratta@iitd.ac.in
Dr. Srivatsava Naidu: Faculty, Center for Biomedical Engineering, IIT Ropar, srivatsava.naidu@iitrpr.ac.in
Dr. Lalitesh Kumar: Project Supervisor and Co-author, Khalifa University, 11932075@zju.edu.cn